

ENGINEERING MECHANICS

UNIT III: BEAMS AND TRUSSES

Harcourt Butler Technical University (HBTU), Kanpur — B.Tech 1st Year

Internal Force Evaluation, Equilibrium Differential Equations, SFD/BMD Profiles, Method of Joints, and
Method of Sections

Contents

1	Part 1: Beams	1
1.1	1.1 Introduction to Beams	1
1.1.1	1.1.1 Classification of Supports in 2D	1
1.1.2	1.1.2 Classification of Beams	1
1.1.3	1.1.3 Types of Transverse Loading	2
1.2	1.2 Internal Forces: Shear Force and Bending Moment	2
1.2.1	1.2.1 Standard Sign Conventions (Crucial for HBTU Examinations)	2
1.3	1.3 Differential Equations of Equilibrium for Beams	2
1.3.1	1.3.1 Fundamental Diagram Construction Rules	3
1.4	1.4 Standard SFD and BMD Profiles for Determinate Beams	4
2	Part 2: Trusses	4
2.1	2.1 Introduction to Trusses	4
2.1.1	2.1.1 Fundamental Assumptions in Ideal Truss Analysis	4
2.2	2.2 Classification and Determinacy of 2D Trusses	5
2.3	2.3 Zero-Force Members (Inspection Rules)	5
2.4	2.4 Method of Joints	5
2.5	2.5 Method of Sections	6
3	Part 3: Representative Analytical Solved Frameworks	6
3.1	3.1 Problem 1: Overhanging Beam with UDL and Point Load (SFD, BMD, Contraflexure) . .	6
3.2	3.2 Problem 2: Simple Truss Analysis by Method of Joints	7
3.3	3.3 Problem 3: Pratt Truss Analysis by Method of Sections	7
4	Part 4: HBTU Examination Quick Reference & Pitfalls	8

1 Part 1: Beams

1.1 1.1 Introduction to Beams

A **beam** is a structural member subjected to transverse loads that act perpendicular to its longitudinal axis. These transverse forces induce internal shear forces and bending moments along the member.

1.1.1 Classification of Supports in 2D

- **Roller Support:** Prevents translation perpendicular to the supporting plane; allows horizontal translation and rotation. Number of independent reactions = 1 (perpendicular to surface).
- **Hinged / Pinned Support:** Prevents translation in both horizontal (X) and vertical (Y) directions; allows free rotation. Number of independent reactions = 2 (R_x, R_y).

- **Fixed / Built-in Support (Cantilever):** Completely constrains translation in all directions and prevents rotation. Number of independent reactions = 3 (R_x , R_y , and fixing moment M_R).

1.1.2 Classification of Beams

A. Statically Determinate Beams (Can be fully solved using the 3 equations of static equilibrium: $\sum F_x = 0$, $\sum F_y = 0$, $\sum M = 0$):

1. **Simply Supported Beam:** Supported by a hinge at one end and a roller at the other ($2 + 1 = 3$ reactions).
2. **Cantilever Beam:** Fixed rigidly at one end and completely free at the other (3 reactions at fixed base).
3. **Overhanging Beam:** Supported by a hinge and roller, where one or both ends extend beyond the supports.

B. Statically Indeterminate Beams (Number of reactions exceeds equations of static equilibrium):

1. **Propped Cantilever Beam:** Fixed at one end and supported on a roller at the other ($3 + 1 = 4$ reactions; indeterminacy = 1).
2. **Fixed Beam:** Built-in rigidly at both ends ($3 + 3 = 6$ reactions; indeterminacy = 3).
3. **Continuous Beam:** Supported over more than two independent supports.

1.1.3 Types of Transverse Loading

- **Concentrated / Point Load (P):** Acts at a discrete point on the beam axis (measured in N or kN).
- **Uniformly Distributed Load (UDL, w):** Constant intensity distributed per unit length (measured in N/m or kN/m). Total equivalent force = $w \cdot L$ acting at the midpoint $L/2$.
- **Uniformly Varying Load (UVL):** Linearly varying intensity from zero to peak w_0 (triangular or trapezoidal). For a triangular load over span L , equivalent force = $\frac{1}{2}w_0L$ acting at centroid ($L/3$ from the heavy base).
- **Concentrated Moment (M_0):** External pure twisting/bending couple applied at a localized point (measured in N · m or kN · m).

1.2 Internal Forces: Shear Force and Bending Moment

When a beam is imagined to be cut at any arbitrary transverse cross-section $X - X$:

Definitions

- **Shear Force (V):** The algebraic sum of all unbalanced external transverse (vertical) forces acting either to the left or to the right of the cross-section:

$$V(x) = \sum F_{y,\text{left}} = - \sum F_{y,\text{right}} \quad (1)$$

- **Bending Moment (M):** The algebraic sum of moments of all external forces and couples acting to the left or to the right of the cross-section taken about the centroid of that section:

$$M(x) = \sum M_{\text{left}} = - \sum M_{\text{right}} \quad (2)$$

1.2.1 Standard Sign Conventions (Crucial for HBTU Examinations)

- **Shear Force Sign Convention:**
 - **Positive (+ V):** Left segment pushes **UPWARD** ($\uparrow$) and Right segment pushes **DOWNWARD** ($\downarrow$). (Clockwise shearing tendency).

- **Negative ($-V$):** Left segment pushes **DOWNWARD** ($\downarrow$) and Right segment pushes **UPWARD** ($\uparrow$).
- **Bending Moment Sign Convention:**
 - **Sagging (Positive, $+M$):** Bends the beam concave upwards ("smile" shape). Causes compression in top fibers and tension in bottom fibers. Left-clockwise and Right-counter-clockwise moments.
 - **Hogging (Negative, $-M$):** Bends the beam convex upwards ("frown" shape). Causes tension in top fibers and compression in bottom fibers. Typical over supports of overhanging beams and throughout cantilevers.

1.3 Differential Equations of Equilibrium for Beams

Consider an infinitesimal element of length dx isolated from a beam subjected to a distributed load of intensity $w(x)$ acting downwards.

- At face x : Shear force is V , Bending moment is M .
- At face $x + dx$: Shear force is $V + dV$, Bending moment is $M + dM$.
- Transverse downward load on element: $dF = w(x) dx$.

Derivation of Differential Relationships

1. Vertical Force Equilibrium ($\sum F_y = 0$):

$$V - w(x) dx - (V + dV) = 0 \implies \frac{dV}{dx} = -w(x) \quad (3)$$

Physical Law: The slope of the Shear Force Diagram (SFD) at any cross-section is equal to the negative of the distributed load intensity at that section. Integrating between points A and B :

$$V_B - V_A = - \int_{x_A}^{x_B} w(x) dx = -(\text{Area under the load diagram}) \quad (4)$$

2. Moment Equilibrium ($\sum M_O = 0$ about right face $x + dx$):

$$-M + (M + dM) - V dx + w(x) dx \left(\frac{dx}{2} \right) = 0 \quad (5)$$

Neglecting higher-order infinitesimals $(dx)^2 \rightarrow 0$:

$$\frac{dM}{dx} = V(x) \quad (6)$$

Physical Law: The slope of the Bending Moment Diagram (BMD) at any cross-section is equal to the shear force at that section. Integrating between points A and B :

$$M_B - M_A = \int_{x_A}^{x_B} V(x) dx = \text{Area under the Shear Force Diagram (SFD)} \quad (7)$$

3. Combined Differential Equation:

$$\frac{d^2M}{dx^2} = \frac{dV}{dx} = -w(x) \quad (8)$$

1.3.1 Fundamental Diagram Construction Rules

1. **Maximum / Minimum Bending Moment:** Setting $\frac{dM}{dx} = V(x) = 0$. The bending moment attains an extreme value (maximum or minimum) at sections where the **shear force is zero or changes sign**.

- 2. Point of Contraflexure (Point of Inflection):** A cross-section where the **bending moment is zero and changes its sign** from positive (sagging) to negative (hogging) or vice-versa. At this point, the beam curvature changes sense, and bending stresses vanish.
- 3. Effect of Concentrated Load:** Produces an abrupt, vertical step change in the SFD equal to the magnitude of the load. No discontinuity in the BMD, but a sharp corner (slope discontinuity).
- 4. Effect of Concentrated Moment (M_0):** Produces an abrupt, vertical step change in the BMD equal to M_0 . No effect on the SFD.

Span Loading Condition	Load Intensity $w(x)$	SFD Profile Shape $V(x)$	BMD Profile Shape
No Load (Between Point Loads)	$w = 0$	Horizontal Line (Constant, 0-degree)	Inclined Straight Line
Uniformly Distributed (UDL)	$w = \text{constant}$	Inclined Straight Line (1st degree)	Parabolic Curve (2nd degree)
Uniformly Varying (UVL)	$w(x) = kx$	Parabolic Curve (2nd degree)	Cubic Parabola (3rd degree)

Table 1: Mathematical Relationship and Profiles of Load, SFD, and BMD

1.4 Standard SFD and BMD Profiles for Determinate Beams

1. Cantilever with Free-End Point Load (P , span L):

- Reactions: $R_A = P \uparrow$, $M_A = -PL$.
- Shear Force: $V(x) = +P$ (uniform rectangle throughout).
- Bending Moment: $M(x) = -P(L - x)$ (linear triangle; maximum hogging $-PL$ at fixed base, zero at free end).

2. Cantilever with UDL (w , span L):

- Reactions: $R_A = wL \uparrow$, $M_A = -\frac{wL^2}{2}$.
- Shear Force: $V(x) = +w(L - x)$ (linear triangle; wL at base to 0 at free end).
- Bending Moment: $M(x) = -\frac{w(L-x)^2}{2}$ (2nd-degree parabola; maximum hogging $-\frac{wL^2}{2}$ at fixed base).

3. Simply Supported Beam with Central Concentrated Load (P at midspan $L/2$):

- Reactions: $R_A = R_B = P/2$.
- Shear Force: $+P/2$ from $x = 0$ to $L/2$, steps down to $-P/2$ from $x = L/2$ to L .
- Bending Moment: Linear increase to $M_{\max} = +\frac{PL}{4}$ at center, decreasing linearly to 0 at ends.

4. Simply Supported Beam with Entire Span UDL (w):

- Reactions: $R_A = R_B = \frac{wL}{2}$.
- Shear Force: $V(x) = \frac{wL}{2} - wx$ (sloping straight line; $+wL/2$ at A , 0 at $x = L/2$, $-wL/2$ at B).
- Bending Moment: $M(x) = \frac{wLx}{2} - \frac{wx^2}{2}$ (symmetric parabolic curve; $M_{\max} = +\frac{wL^2}{8}$ at midspan).

2 Part 2: Trusses

2.1 Introduction to Trusses

A **truss** is a triangulated structure composed of slender, straight members interconnected at their ends by pin joints (frictionless hinges). Trusses are engineered to support major spans in bridges, roof frameworks, and transmission towers with optimal weight efficiency.

2.1.1 Fundamental Assumptions in Ideal Truss Analysis

1. All members are straight and joined at their ends by frictionless smooth pins.
2. Loads and support reactions are applied strictly and exclusively at the joints (nodes).
3. The longitudinal centroidal axes of all members concurring at a joint are concurrent at a single point.
4. The self-weight of individual members is either negligibly small or divided equally between the two bounding joints.
5. Consequently, all members act as **two-force members**, carrying exclusively **axial forces** (pure tension or pure compression); bending moments and shear forces inside members are zero.

2.2 2.2 Classification and Determinacy of 2D Trusses

For a plane truss with m members and j joints:

- At each joint, 2 independent equilibrium equations exist ($\sum F_x = 0, \sum F_y = 0$).
- Total available joint equations = $2j$.
- Unknowns to determine = m member axial forces + 3 support reactions = $m + 3$.

Determinacy Criterion

- **Perfect / Statically Determinate Truss:**

$$m = 2j - 3 \quad (9)$$

The structure is geometrically rigid, stable, and all internal member forces can be solved uniquely using equations of statics.

- **Deficient / Imperfect Truss ($m < 2j - 3$):** The structure possesses fewer members than required for rigidity. It acts as an unstable mechanism and collapses under general loads.
- **Redundant / Statically Indeterminate Truss ($m > 2j - 3$):** The structure contains extra members. The equations of statics are insufficient; compatibility conditions involving elastic deformation must be used.

2.3 2.3 Zero-Force Members (Inspection Rules)

Before performing detailed calculations, members carrying zero axial force can be rapidly identified by visual inspection:

Two Critical Zero-Force Rules

Case 1: Two Non-Collinear Members at an Unloaded Joint If two members meet at a joint that is subjected to **no external load and no support reaction**, both members must carry **zero force**:

$$F_1 = 0 \quad \text{and} \quad F_2 = 0 \quad (10)$$

Case 2: Three Members at an Unloaded Joint (Two Collinear) If three members meet at an unloaded joint where two of the members are **collinear**, the third non-collinear member must carry **zero force**:

$$F_{\text{non-collinear}} = 0, \quad F_{\text{collinear 1}} = F_{\text{collinear 2}} \quad (11)$$

2.4 2.4 Method of Joints

- **Philosophy:** Since the entire truss is in static equilibrium, every single individual joint pin must independently be in equilibrium under the concurrent forces acting on it.

- **Governing Equations at Each Joint:**

$$\sum F_x = 0, \quad \sum F_y = 0 \quad (12)$$

- **Analytical Workflow:**

1. Calculate external support reactions using global equilibrium ($\sum F_x = 0, \sum F_y = 0, \sum M = 0$).
2. Identify an initial joint with **at most two unknown member forces**.
3. Draw the FBD of the joint. Standard practice is to assume all unknown member forces are in **Tension** (pulling away from the joint).
4. Apply $\sum F_x = 0$ and $\sum F_y = 0$ to solve for the forces.
5. If the calculated value is positive, the member is in **Tension** (T); if negative, the member is in **Compression** (C).
6. Move systematically to adjacent joints with now no more than two remaining unknowns.

- **Best Suited For:** Complete truss analysis where forces in **all** members are required.

2.5 Method of Sections

- **Philosophy:** An imaginary cutting plane (section line) is passed completely through the truss, cutting members of interest and dividing the truss into two completely separated parts. Either isolated part is treated as a rigid body in static equilibrium.

- **Governing Equations for Isolated Segment:**

$$\sum F_x = 0, \quad \sum F_y = 0, \quad \sum M_O = 0 \quad (\text{about any strategic joint } O) \quad (13)$$

- **Crucial Rule of Section Cutting:** The section line must not cut more than **three non-concurrent members** whose forces are unknown (since there are only 3 independent 2D equilibrium equations).

- **Analytical Workflow:**

1. Find external support reactions.
2. Pass an imaginary cut through the members whose forces are sought (maximum 3 unknowns).
3. Select the simpler side of the cut truss and draw its complete FBD, replacing cut members with internal axial forces directed away from the cut (assumed tension).
4. Select a strategic moment pivot joint O where two of the three unknown lines of action intersect. Taking $\sum M_O = 0$ yields a single equation with one unknown force directly!
5. Use $\sum F_y = 0$ or $\sum F_x = 0$ to solve the remaining forces.

- **Best Suited For:** Rapidly finding forces in specific selected interior or chord members without solving joint-by-joint from the supports.

3 Part 3: Representative Analytical Solved Frameworks

3.1 Problem 1: Overhanging Beam with UDL and Point Load (SFD, BMD, Contraflexure)

Problem Statement: A beam $ABCD$ is simply supported at B and C with an overhang $AB = 2$ m, span $BC = 6$ m, and overhang $CD = 2$ m (total length 10 m). It carries a point load of 10 kN at free end A , a UDL of 4 kN/m over the entire span BC , and a point load of 6 kN at free end D . Determine support reactions, draw SFD and BMD, find maximum positive bending moment, and locate points of contraflexure.

Step-by-step Solution:

1. **Support Reactions:** Let vertical reactions at B and C be R_B and R_C :

$$\begin{aligned}\sum M_B = 0 &\implies -(10 \times 2) + (4 \times 6) \left(\frac{6}{2}\right) - R_C(6) + 6 \times (6 + 2) = 0 \\ &\implies -20 + (24)(3) - 6R_C + 48 = 0 \implies -20 + 72 + 48 - 6R_C = 0 \\ 100 - 6R_C = 0 &\implies \mathbf{R_C = 16.67 \text{ kN}}\end{aligned}$$

$$\text{Vertical equilibrium: } \sum F_y = 0 \implies R_B + R_C = 10 + (4 \times 6) + 6 = 40 \text{ kN}$$

$$\mathbf{R_B = 40 - 16.67 = 23.33 \text{ kN}}$$

2. **Shear Force Calculations:**

- Just right of A : $V_A = -10 \text{ kN}$ (constant from A to left of B).
- Just left of B : $V_{B-} = -10 \text{ kN}$.
- Just right of B : $V_{B+} = -10 + R_B = -10 + 23.33 = +13.33 \text{ kN}$.
- In span BC (x from B): $V(x) = +13.33 - 4x$.
- Point of Zero Shear: Set $V(x) = 0 \implies 13.33 - 4x = 0 \implies \mathbf{x_0 = 3.33 \text{ m}}$ from support B .
- Just left of C : $V_{C-} = 13.33 - 4(6) = 13.33 - 24 = -10.67 \text{ kN}$.
- Just right of C : $V_{C+} = -10.67 + R_C = -10.67 + 16.67 = +6.0 \text{ kN}$.
- Overhang CD : Constant $V = +6 \text{ kN}$ until end D , where load drops it to zero ($6 - 6 = 0$).

3. **Bending Moment Calculations:**

- At free end A : $M_A = 0$.
- At support B : $M_B = -10 \times 2 = \mathbf{-20 \text{ kN} \cdot \text{m}}$ (hogging).
- At support C : $M_C = -6 \times 2 = \mathbf{-12 \text{ kN} \cdot \text{m}}$ (hogging).
- At free end D : $M_D = 0$.
- Maximum Positive Bending Moment (at $x_0 = 3.33 \text{ m}$ from B):

$$\begin{aligned}M_{\max} &= M_B + V_{B+}(x_0) - \frac{wx_0^2}{2} \\ &= -20 + (13.33)(3.33) - \frac{4(3.33)^2}{2} = -20 + 44.42 - 22.21 = \mathbf{+2.21 \text{ kN} \cdot \text{m}} \quad (\text{sagging})\end{aligned}$$

4. **Points of Contraflexure** ($M(x) = 0$ in span BC):

$$M(x) = -20 + 13.33x - 2x^2 = 0 \implies 2x^2 - 13.33x + 20 = 0$$

$$\text{Solving using quadratic formula } x = \frac{13.33 \pm \sqrt{(-13.33)^2 - 4(2)(20)}}{4} = \frac{13.33 \pm \sqrt{177.69 - 160}}{4} = \frac{13.33 \pm \sqrt{17.69}}{4};$$

$$\mathbf{x_1 = \frac{13.33 - 4.21}{4} = 2.28 \text{ m},} \quad \mathbf{x_2 = \frac{13.33 + 4.21}{4} = 4.38 \text{ m}} \quad (\text{both measured from support } B)$$

Thus, there are **two points of contraflexure** located at 2.28 m and 4.38 m to the right of support B .

3.2 Problem 2: Simple Truss Analysis by Method of Joints

Problem Statement: An equilateral triangular truss ABC of span $L = 4 \text{ m}$ consists of members AB, BC, CA each of length 4 m ($\angle A = \angle B = \angle C = 60^\circ$). It is hinged at A and supported on rollers at B . A vertical downward load of 20 kN is applied at apex C . Determine all member forces.

Step-by-step Solution:

1. **Reactions:** By symmetry:

$$R_A = R_B = \frac{20}{2} = 10 \text{ kN } \uparrow, \quad H_A = 0$$

2. **Joint A Equilibrium:** Let F_{AC} and F_{AB} be tensions (pulling away from joint A):

$$\begin{aligned}\sum F_y = 0 &\implies R_A + F_{AC} \sin 60^\circ = 0 \implies 10 + F_{AC} \left(\frac{\sqrt{3}}{2} \right) = 0 \\ &\implies \mathbf{F_{AC} = -\frac{20}{\sqrt{3}} \approx -11.55 \text{ kN} \implies 11.55 \text{ kN (Compression)}} \\ \sum F_x = 0 &\implies F_{AB} + F_{AC} \cos 60^\circ = 0 \implies F_{AB} + (-11.55)(0.5) = 0 \\ &\implies \mathbf{F_{AB} = +5.77 \text{ kN (Tension)}}\end{aligned}$$

3. **Joint B Equilibrium:** By bilateral symmetry: $\mathbf{F_{BC} = 11.55 \text{ kN (Compression)}}$.

3.3 Problem 3: Pratt Truss Analysis by Method of Sections

Problem Statement: A simply supported Pratt truss with panel lengths of 3 m and height 4 m carries vertical joint loads. A section cuts top chord member CD , diagonal member KD , and bottom chord member KL . Formulate the equilibrium equations to solve member forces directly.

Step-by-step Solution:

1. **Cut Plane:** Pass vertical section line cutting through CD , KD , and KL . Keep left isolated segment.
2. **Solve Bottom Chord Force F_{KL} :** Take moments about opposite top joint D where CD and KD intersect:

$$\sum M_D = 0 \implies (R_A \cdot x_D) - \sum (P_i \cdot x_i) - F_{KL} \cdot h = 0 \implies \mathbf{F_{KL} = \frac{\sum M_{\text{external}}}{h}}$$

3. **Solve Top Chord Force F_{CD} :** Take moments about joint K where KD and KL intersect:

$$\sum M_K = 0 \implies \mathbf{F_{CD} = \frac{\sum M_K}{h}}$$

4. **Solve Diagonal Force F_{KD} :** Apply vertical force equilibrium on the section:

$$\sum F_y = 0 \implies R_A - \sum P_{\text{left}} \pm F_{KD} \sin \theta = 0 \implies \mathbf{F_{KD} = \frac{V_{\text{panel}}}{\sin \theta}}$$

4 Part 4: HBTU Examination Quick Reference & Pitfalls

High-Yield HBTU Examination Key Takeaways

- **Parabola Curvature in BMD:** Under downward UDL, the BMD curve is **hogging/sagging concave downward** (differentiating: $\frac{d^2 M}{dx^2} = -w < 0$, meaning a local maximum with downward concavity).
- **Point of Contraflexure Condition:** Do not confuse $V = 0$ with $M = 0$. $V = 0$ yields M_{max} , whereas $M = 0$ (with sign change) identifies the **point of contraflexure**.
- **Section Cut Rule in Trusses:** Never pass a section line that cuts more than 3 members with unknown forces unless some are proven zero-force members beforehand.
- **Sign Presentation in Truss Tables:** In exam answers, always explicitly append **(T)** for Tension or **(C)** for Compression alongside the numerical value; never leave raw negative/positive signs without clarifying the physical mode.